geolens

Valley and channel geomorphology from a DEM — what it does, how it works, and how far to trust it

Version	0.28.0 (September 2026)
Licence	MIT; built entirely on open-source libraries (GDAL, rasterio, geopandas, shapely, pysheds). No ArcGIS licence is needed for anything.
Built for	Lidar DEMs of arid and semi-arid Southwest watersheds, including state-plane grids in US survey feet (e.g. EPSG:6529, NM Central)
Ways in	Jupyter notebook (*geolens_start.ipynb*), Python API (*geolens.api*), command line (*geolens …*), and an MCP server so Claude can run it

1 Summary

geolens takes a digital elevation model and returns a reach-by-reach description of the valleys and channels on it, as GIS layers you can open in QGIS or ArcGIS. One call — *api.landscape_pass* — conditions the DEM and routes flow, extracts a stream network and the height above it (HAND), maps valley bottoms, splits the network into reaches and classes their confinement, measures each valley from its floor to its divides, and assigns two Rosgen classifications: **valley types I–XI** on the valley, and **stream types A–G** on the channel, measured from cross-sections cut on the finest DEM available — and then judges whether each reach's stream type is a **stable or unstable form** for its valley type. Separate calls map gullies, headcuts and alluvial fans, fetch surficial geology and NAIP imagery, and map a riparian corridor from NDVI.

The design rule throughout is that a classification says what it rests on. Every valley and stream type carries a confidence and a basis field, candidates are listed when the data cannot separate them, and measurements that the grid cannot resolve are reported as unknown rather than filled with a guess. Section 8 sets out what has been verified and what has not.

The stream-type workflow was rebuilt in versions 0.24–0.28. On the Eagle Creek test area (0.61 m lidar) v0.23 left 33 of 53 reaches without a stream type and discarded 49 of its 61 bankfull picks as implausible; v0.25, replayed on the same cross-sections, typed all 53 (Section 7). Checked against Moody et al.'s (2003) New Mexico field data on the full Ruidoso run, that bank picker still ran 2–2.6× too deep in small drainages, so since v0.28 bankfull is set at the regional curve's depth and the DEM supplies the shape (Section 5.2). Every cross-section also has a plot, linked to its line in QGIS or ArcGIS.

2 Inputs, workflow and outputs

Input	Required?	Used for
DEM	yes	Flow routing, HAND, valley bottoms, valley form. A 2–10 m grid is usually enough for routing.
DEM_FINE	no	Channel cross-sections for stream types only. Point samples along each section, so a multi-gigabyte 1 m lidar DEM is never loaded or resampled.
Stream network (NHD)	no	Joins NHD identity to the derived reaches.
Geology	no ("auto")	Material per reach from the USGS Cooperative National Geologic Map; separates valley types that shape cannot, and gives the stream-type substrate number as a prior.
Regional curve	no	Expected bankfull width and depth from drainage area (Section 5.3). *arid_sw_nm* for New Mexico and eastern Arizona.
Centerline	no	Reach-scale cross-sections and bank delineation for a single surveyed reach.

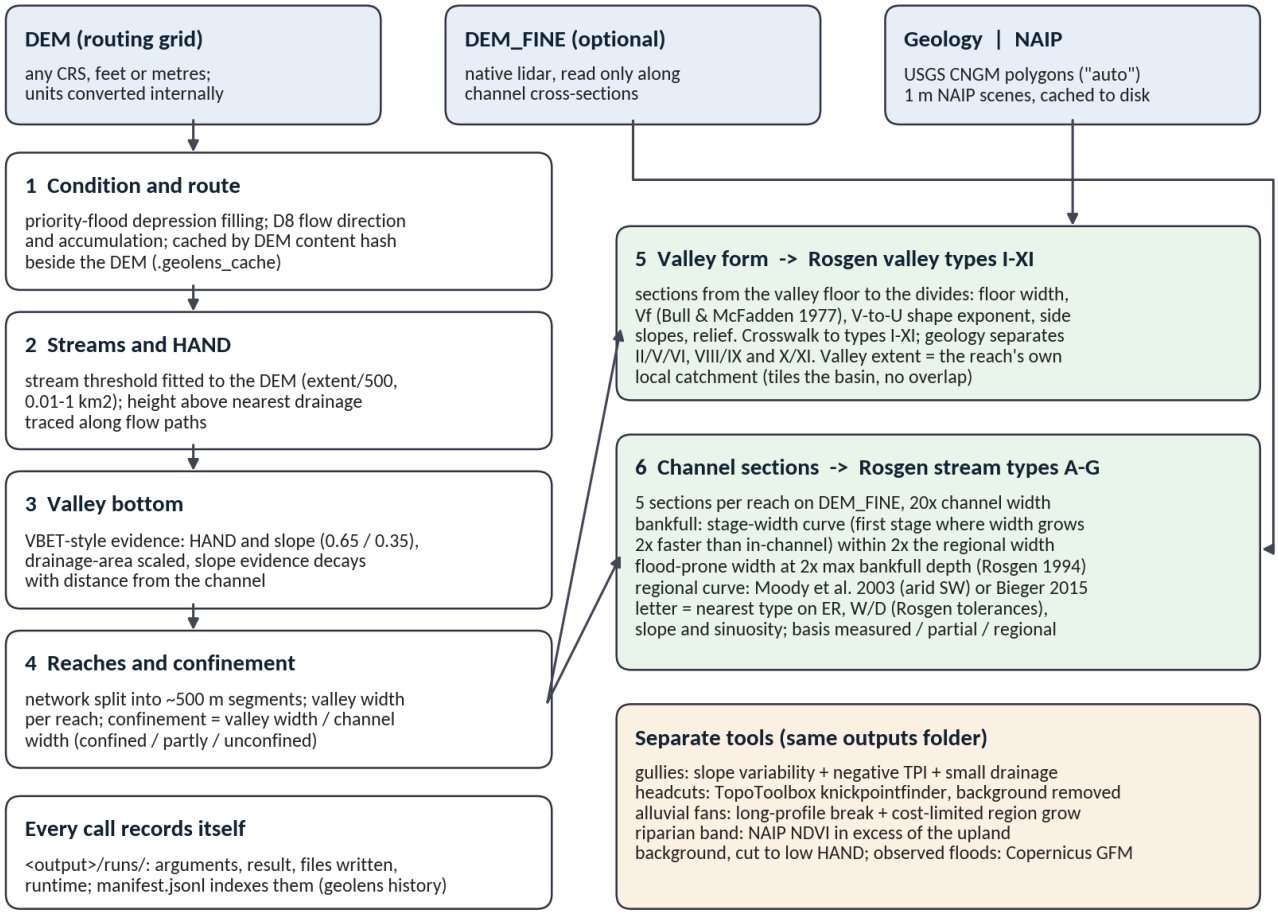


Figure 1. The landscape pass (steps 1–6) and the separate tools. Routing runs once per DEM and is cached beside it, so later runs with different thresholds skip the expensive part.

Every result goes under the output folder (*GEOLENS_OUT*), one sub-folder per step. The layers most people open first:

Folder / file	What it holds
network_dir/segments.gpkg	The stream network split into reaches (default 500 m).
classified_dir/segments_classified.gpkg	Per reach: drainage area, gradient, valley width, estimated channel width, confinement ratio and class.
valley_dir/valley_bottom.gpkg	Valley-bottom polygons (the low-lying inner core is in valley_core.gpkg).
valley_types_dir/valley_types.gpkg	Rosgen valley type, candidates, confidence and basis per reach.
valley_extent_dir/valley_extent.gpkg	Each reach's valley as a polygon: its local catchment, bounded by the divides.
stream_types_dir/stream_types.gpkg	Rosgen stream type per reach with entrenchment, width/depth, sinuosity, slope, basis and confidence.
stream_types_dir/stream_type_sections.gpkg, stream_type_banks.gpkg, stream_type_profiles.csv	Every channel cross-section, the bank-top and flood-prone points picked on it, and the profile itself — the evidence behind each stream type.
runs/	A record of every call: arguments, result, files written and run time, indexed in *manifest.jsonl*.

3 Terrain, streams and valley bottoms

3.1 Conditioning and flow routing

Depressions are filled with the priority-flood algorithm (Barnes et al., 2014), flow directions are D8 (O'Callaghan and Mark, 1984) and flow accumulation gives drainage area. pysheds does the routing; WhiteboxTools (Lindsay, 2016) is an optional backend that *backend="auto"* prefers above about 50 million cells when it is installed, and it has not yet been verified against pysheds here (*geolens compare-backends*). Because routing dominates the run time on a large DEM, its outputs are cached in a *.geolens_cache* folder beside the DEM, keyed on the DEM's path and size; when its file date changes, a hash of its content decides whether the cache is still valid, so copying or touching the file does not throw hours of routing away. Any later run on the same DEM — with a different stream threshold, say — reuses it. *plan_landscape_run* reads only the DEM header and reports cells and peak memory before anything is started.

3.2 Stream network and HAND

The channel-initiation threshold is fitted to the DEM unless one is given: its extent divided by 500, kept between 0.01 and 1 km^2. A fixed 1 km^2 threshold on a small DEM returns only the main stems. Height above nearest drainage (HAND; Rennó et al., 2008; Nobre et al., 2011) is traced along the flow paths to the stream cell each cell drains to, after restricting stream cells to valid ground so that flow collecting in a clipped DEM's nodata collar cannot become a stream.

3.3 Valley bottoms

Valley bottoms follow the Valley Bottom Extraction Tool approach (VBET; Gilbert et al., 2016): a HAND evidence surface and a slope evidence surface are combined as $E = E_{HAND}^{0.65} \times E_{slope}^{0.35}$. The HAND threshold scales with drainage area, so one parameter set bounds a headwater floor and a river's; slope evidence is down-weighted with distance from the channel, which is what stops flat uplands — mesas, terraces, ridge tops — reading as valley floor. VBET itself is GPL-3.0 and cannot be bundled in an MIT package, so this ~100-line evidence layer is implemented here.

3.4 Reaches and confinement

The network is split into reaches (500 m by default). For each, valley width is the valley-bottom area allocated to it divided by its length, and channel width comes from a drainage-area relation ($W = 2.7 \cdot A^{0.35}$, m and km^2; Bieger et al., 2015) unless a width is supplied. Their ratio is the confinement ratio (cf. Fryirs et al., 2016); reaches are confined below 2, partly confined from 2 to 5 and unconfined above 5 by default. Those breaks are not calibrated to any landscape: *confinement_break_sweep* compares candidate breaks against the ratio distribution and *reclassify_confinement* applies new ones in seconds, without re-routing.

4 Valley form and Rosgen valley types

Rosgen's valley types (Rosgen, 1996) are defined by the whole valley — V-notched canyons, gentle side slopes, U-shaped troughs, broad alluvial floors — so geolens measures the valley as a landform. For each reach, long sections are cut across the valley and walked outward and upward from the floor to the divide on each side. Each section gives the valley floor width, the valley-floor width to height ratio $V_f = 2V_{fw} / [(E_{ld} - E_{sc}) + (E_{rd} - E_{sc})]$ (Bull and McFadden, 1977), the exponent n of $z \sim |x|^n$ fitted above the floor (n ≈ 1 a V, n ≈ 2 a U), side slopes from floor edge to divide, relief, and asymmetry.

A crosswalk maps those measurements, with material from the geology layer and, when supplied, detected alluvial fans, to types I–XI. Several pairs of types differ by depositional history rather than shape (II/V/VI, VIII/IX, X/XI); without geology they come back unresolved with their candidates listed, and type VI (structurally controlled) is never asserted, because structure is not visible in a DEM or a geologic map. Each reach's valley extent is its own local catchment: the area draining to that reach, bounded by the divides, so the extents tile the basin without overlapping.

Geology comes from the USGS Cooperative National Geologic Map (CNGM; U.S. Geological Survey, 2025), fetched with *geology="auto"* from its public feature services and joined to a small vocabulary of materials (alluvial, colluvial, bedrock, lacustrine and so on). The source maps are 1:50,000 to 1:1,000,000, so a confined valley tens of metres wide often returns its hillslope's bedrock unit. *unit_area_ratio* on each polygon, a warning when one unit covers most of the area, and the per-reach material cover are there to catch that.

5 Rosgen stream types

Stream types A–G describe the channel, not the valley, and Rosgen's Level II key (Rosgen, 1994, 1996) discriminates them with four measurements: the **entrenchment ratio** (flood-prone width divided by bankfull width, the flood-prone width being taken at twice the maximum bankfull depth), the **width/depth ratio** (bankfull width over mean depth), **sinuosity** and **slope**. Everything here turns on finding bankfull on a DEM, which is where the work of versions 0.24–0.28 went.

5.1 Channel cross-sections

Five sections are cut across each reach, perpendicular to the channel and 20 times the classifier's channel width long (at least 60 m), so that they reach past the flood-prone width. They are sampled on DEM_FINE when given, at the cell size but no finer than 0.5 m. Offsets and elevations are converted to metres, each with its own factor, before anything is measured; a grid in US survey feet with elevations in feet or in metres gives the same answer, and a test feeds the same profiles through in both units to prove it.

5.2 Bankfull at the regional depth (default since v0.28)

At each section the water is raised from the thalweg to the maximum bankfull depth the regional curve gives for the reach's drainage area (Moody et al.'s mean depth / 0.58), and the width, area, mean depth and flood-prone width (at twice that depth) are read off the DEM at that stage. Where that depth would spill a small channel onto a flat floor — a width over three times the regional one — the stage is lowered until the width is plausible, but not below 0.1 m, about the vertical noise of 1 m lidar. A section with nothing channel-shaped at that scale falls back to the regional entrenchment. Because the depth comes from the curve, no reach typed this way is rated above medium confidence.

Why the change. The bank picker below was checked against Moody et al.'s New Mexico field data on Sam's Ruidoso run (1.785 billion cells at 0.76 m; 1,875 usable picks). Its depths ran a median 2.1–2.6× the field depth in drainages under 20 km^2, only one pick in four or five within a factor of 1.5: at this resolution the first break it can see is usually the top of a small incised channel, or in a wide sandy wash a low-flow thread, and a too-deep pick also inflates the entrenchment ratio. Replayed on the same sections, the regional stage gave a usable bankfull on 3,336 sections instead of 1,875, a median width 1.33× the regional width instead of 1.71×, and a median width/depth of 27 against the curve's 21–24 (the picker: 14.5). The types move a lot: C and E fall from 380 reaches to 55, B, D and F rise — wide sandy washes read as braided D, small incised channels as B. That is a consistent, curve-dependent answer in place of an inconsistent one; it has not been compared with surveyed bankfull on these areas.

5.3 The bank picker (stage–width), now a check

The picker still runs on every section and its answer is kept (*picker_** columns, purple on the section plots, *picker_bank_top* points), so the two can be compared reach by reach; *bankfull_method="picker"* restores it and *"hybrid"* uses it where its depth is within 2× of the regional one. How it works:

Water is raised from the thalweg in 2 cm steps and the connected width W is measured at each stage h, walking out from the thalweg on each side until the ground reaches the water. Inside the channel, width grows slowly because the banks are steep. At bankfull it starts to grow much faster — the water has left the channel, onto a floodplain, a bench or a gentler valley wall. This is Riley's (1972) bench index, one of the width–stage methods compared by Williams (1978) and Navratil et al. (2006). The implementation:

- The thalweg is the lowest sample within one expected width of the mapped channel line, and the search is confined to twice the expected width either side of it, which keeps most valley walls out of reach.
- dW/dh is smoothed over three stages. A **candidate** is any stage at least the minimum bank height up (half the regional maximum depth, between 0.15 and 0.3 m) where dW/dh stays at least **twice** its median since the previous candidate for three consecutive steps. The lowest candidate is bankfull.
- The flood-prone width is read at twice that depth over the whole section; mean depth is the cross-sectional area over the width.
- A pick is not used when it is more than three times the regional width (a bench, terrace or wall), narrower than three samples (the grid, not the channel), or one-sided (the water left the search window on one side). A section with no candidate at all — a plain V — returns "no bank found".

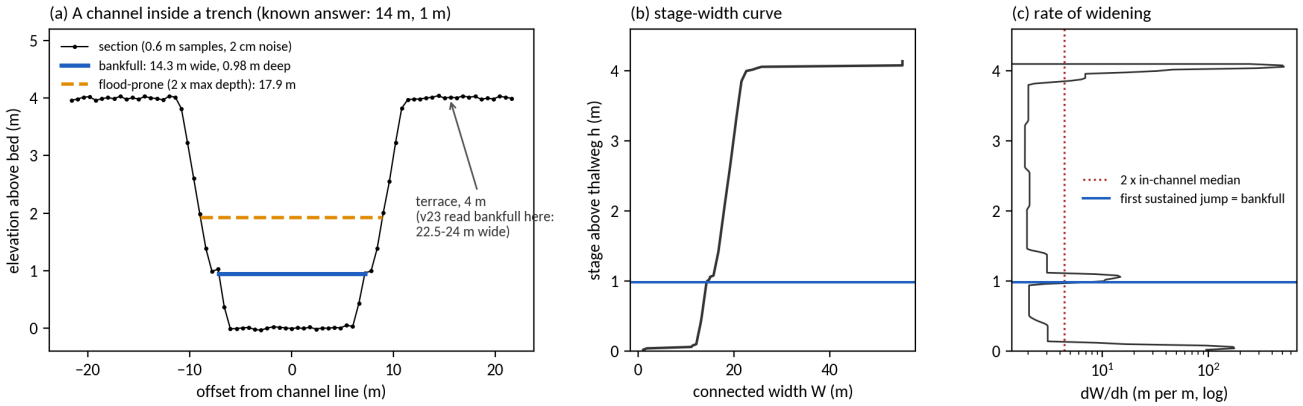


Figure 2. The method on a known-answer profile: a 14 m, 1 m deep channel inside a trench whose walls rise to a terrace at 4 m, sampled at 0.6 m with 2 cm noise. (a) The pick and the flood-prone level. (b) The stage–width curve. (c) The rate of widening; the first sustained jump above twice the in-channel median is bankfull. The previous slope-break method read this channel as 22.5–24 m wide from the terrace.

The method is tested on five synthetic profiles with exact answers — a floodplain channel, a confined channel with no floodplain, the trench above, an asymmetric channel with a floodplain on one side only, and a pure V with no channel — each sampled at 0.25 m without noise and at 0.6 and 1.2 m with 3 cm of noise, and with the expected width deliberately wrong by ×0.6 and ×1.6:

Profile	Truth	Recovered (all spacings, all expected-width errors)
Floodplain	14.0 m × 1.0 m	14.0–14.3 m; depth within 0.04 m
Trench (terrace at 4 m)	14.0 m × 1.0 m	14.0–14.5 m, flood-prone 17.9–18.2 m (truth 18 m)
Asymmetric	8.0 m × 0.8 m	8.0–9.0 m
Pure V	no bankfull	none found, at every spacing
Confined (3:1 banks)	5.0 m × 1.0 m	5.03 m at 0.25 m spacing; none at 0.6 and 1.2 m, where the 0.33 m bank face falls between samples and the sampled profile has no break to find

5.4 Regional curves

A regional curve — bankfull width and mean depth as power laws of drainage area — sizes the bank search, bounds what is plausible, and supplies the channel for reaches whose banks the grid cannot show. The notebook uses the arid-Southwest curves of Moody et al. (2003), converted from feet and square miles:

regional_curve	Width W (m)	Mean depth D (m)	Source
arid_sw_nm	$2.311\ A^{0.2811}$	$0.1134\ A^{0.237}$	Moody et al. (2003), Eastern Arizona / New Mexico: $9.9059\ DA^{0.2811}$ ft, $0.4663\ DA^{0.237}$ ft (DA in mi^2)
arid_sw_az	$3.554\ A^{0.3169}$	$0.1933\ A^{0.219}$	Moody et al. (2003), central and southern Arizona
bieger_us	$2.70\ A^{0.352}$	$0.30\ A^{0.213}$	Bieger et al. (2015), all-US — generic
None (default)	segments' own width, $2.7\ A^{0.35}$	Bieger depth	the classifier's relation

A is drainage area in km^2. Maximum depth is taken as mean depth / 0.58, the riffle relation Moody et al. (2003) report across their Arizona, New Mexico and Navajo Nation sites (R^2 = 0.90). A custom curve can be passed as *{"width": (a, b), "depth": (c, d)}*. Their report includes field sites on an Eagle Creek near Alto, NM — a different place from the Eagle Creek test area in Section 7 — which sit on the New Mexico curve: below the South Fork (8.1 mi^2), 15.8 ft × 0.7 ft measured against 17.8 × 0.77 ft predicted.

5.5 From sections to a reach: the basis

Sections are searched for banks only when the regional width is at least three samples; below that, what the detector finds is the valley floor or a bench, not the channel. A section without a usable pick still carries information: the flood-prone width is several channel widths across and *is* visible, so it is read at twice the regional maximum depth and divided by the regional width. Each reach then records what its letter rests on:

stream_basis	Evidence	Confidence
measured	Usable banks in two or more sections; entrenchment and width/depth are the medians of those sections.	high, medium, or low if flagged *no_type_fits*
partial	Banks in one section only. Its entrenchment is pooled with the regional ones; width/depth is not used.	low
regional	No usable banks. Entrenchment from the flood-prone width against the regional channel; width/depth unknown.	low
slope_sinuosity	No section at all. A letter only where slope and sinuosity separate the types on their own (above 10% only A exists).	low

5.6 The key: the nearest Rosgen type

Rosgen's key is delineative on entrenchment and width/depth; slope and sinuosity are "typical" values, and a reach outside the typical slope keeps its letter with a slope suffix (Bc, Cb, Aa+). Rosgen (1994) also allows a measured entrenchment ratio to sit ±0.2 either side of a boundary and a width/depth ratio ±2.0. geolens therefore scores every type by its **misfit** — how far each measurement lies outside the type's range:

$$\text{misfit} = \Delta ER/0.2 + \Delta(W/D)/2.0 + 0.5 \cdot \log_2(\text{slope distance}) + 0.25 \cdot \Delta\text{sinuosity}/0.1$$

where each Δ is zero inside the range and the slope term is $\log_2$ of the slope over the nearest bound of the typical range. E carries a fixed +0.1, so without a measured width/depth the commoner C wins the C–E tie. Unmeasured criteria add nothing to any type. The type with the least misfit takes the letter when it beats the next by 0.05 (0.25 when entrenchment is unmeasured); otherwise the near-tied types are returned as candidates. Three rules keep the answer inside Rosgen's chart:

- **Slope classes.** One table lists, letter by letter, the slope classes that exist and how they are written: A 4–10%, Aa+ above 10%; Ba, B, Bc; Cb, C, Cc−; Db, D; DA below 0.5%; Eb, E; Fb, F; G, Gc. A letter is excluded at a slope where it does not exist — a 13% reach is never a C — and an entrenched, low width/depth channel is an A above 4% and a G below it.
- **D and DA need what a DEM cannot supply.** D (braided) is offered only when width/depth is measured above 40. DA (anastomosed) is multi-thread, which one centreline cannot show; it can appear among the candidates but never as the letter.
- **Confidence.** High needs entrenchment and width/depth measured, both inside the chosen type's ranges, sinuosity consistent and a clear margin; medium needs both measured and within Rosgen's tolerance of the chosen type; anything else is low, including every call resting on the regional channel. *no_type_fits* flags a measured reach whose entrenchment or width/depth lies beyond that tolerance of the nearest type that exists at its slope — a 13% reach with entrenchment 2.6 is still an Aa+, flagged.

With geology, the substrate number (1 bedrock to 6 silt/clay) is added as a prior from the mapped material — enough to separate a bedrock reach from an alluvial one, not to tell gravel from cobble — giving *stream_type_full* in Rosgen's order, e.g. B4c.

5.7 Stable or unstable form

The point of classifying both the valley and the channel is to ask whether each reach's stream type is a stable form for its valley. The table is NRCS (2007) NEH 654 Technical Supplement 3E, Table TS3E-3, after Rosgen (1996), which lists the stream types associated with each valley type and, for three of them, the types that appear only when the system is out of balance. Each reach is **stable** (its stream type is associated with its valley type), **unstable** (a type listed only under disequilibrium), **check** (not associated with that valley type at all — a departure, or one of the two classifications is wrong) or **unknown**. An unresolved valley type is judged under every candidate. The confidence is the lower of the two classifications'.

Valley type	Stream types	Only out of balance
I	A (Aa+)	
II	B	
III	A, B, G, D	
IV	F	
V	C, D, G	
VI	B, C, F	G (fault disequilibrium)
VII	A, G	
VIII	C, E	D, F, G
IX	C, D	
X	E, C, DA	G, F (base level changed)
XI	DA, C, E	

6 Other tools

Tool	Method	Status
Gullies (*detect_gullies*)	Slope-variability, negative TPI and a drainage-area ceiling (default 0.5 km^2); each gully's width, depth and volume measured on cross-sections. An optional roads layer sets road cuts aside.	6 of 6 on a known-gully fixture, depth −1%. On steep mountain terrain most detections are headwater ravines; footprint polygons (31 m wide, 11 m deep median on Cherokee Bill) are off by default. Roughness runs in strips, so any DEM size fits in a few GB
Headcuts	TopoToolbox knickpointfinder (Schwanghart and Scherler, 2014, 2017) with the background gradient removed	6 of 6, no false positives, 3–6 m
Alluvial fans	Concave break in the channel long profile, then a cost-limited region grow	3 of 3; apex within 7 m, area −5%
NAIP imagery	1 m four-band NAIP scenes streamed from Microsoft Planetary Computer and cached to disk with flight year; NDVI built at 1 m regardless of DEM cell size	Working on Sam's machine (Eagle Creek, 2022 scenes, full coverage)
Riparian band	NDVI in excess of the local upland background (40th percentile in a 2 km window, excess ≥ 0.12), cut to HAND ≤ 15 m; rectangular or disconnected patches flagged as possible agriculture	Synthetic scene only; usefulness under review
Observed floods	Copernicus Global Flood Monitoring archive	For validation; unreliable on ephemeral washes
Reach scale	Sections along a surveyed centerline; bank tops by slope break, floodplain stopped at valley confinement	Floodplain width 0.00% error on a known compound channel

7 The Eagle Creek test area: before and after (v0.25)

The Eagle Creek test area is mapped with 0.61 m lidar in NM State Plane feet. (It is not the Eagle Creek of Moody et al.'s field sites.) These figures show v0.25, before the bankfull change of Section 5.2. The v23 landscape pass on it took about 5 minutes on a desktop and produced 53 reaches. Figure 3 compares its stream types with the current method replayed on the same saved cross-sections with the *arid_sw_nm* curve; Figure 4 shows the banks picked on four of those reaches.

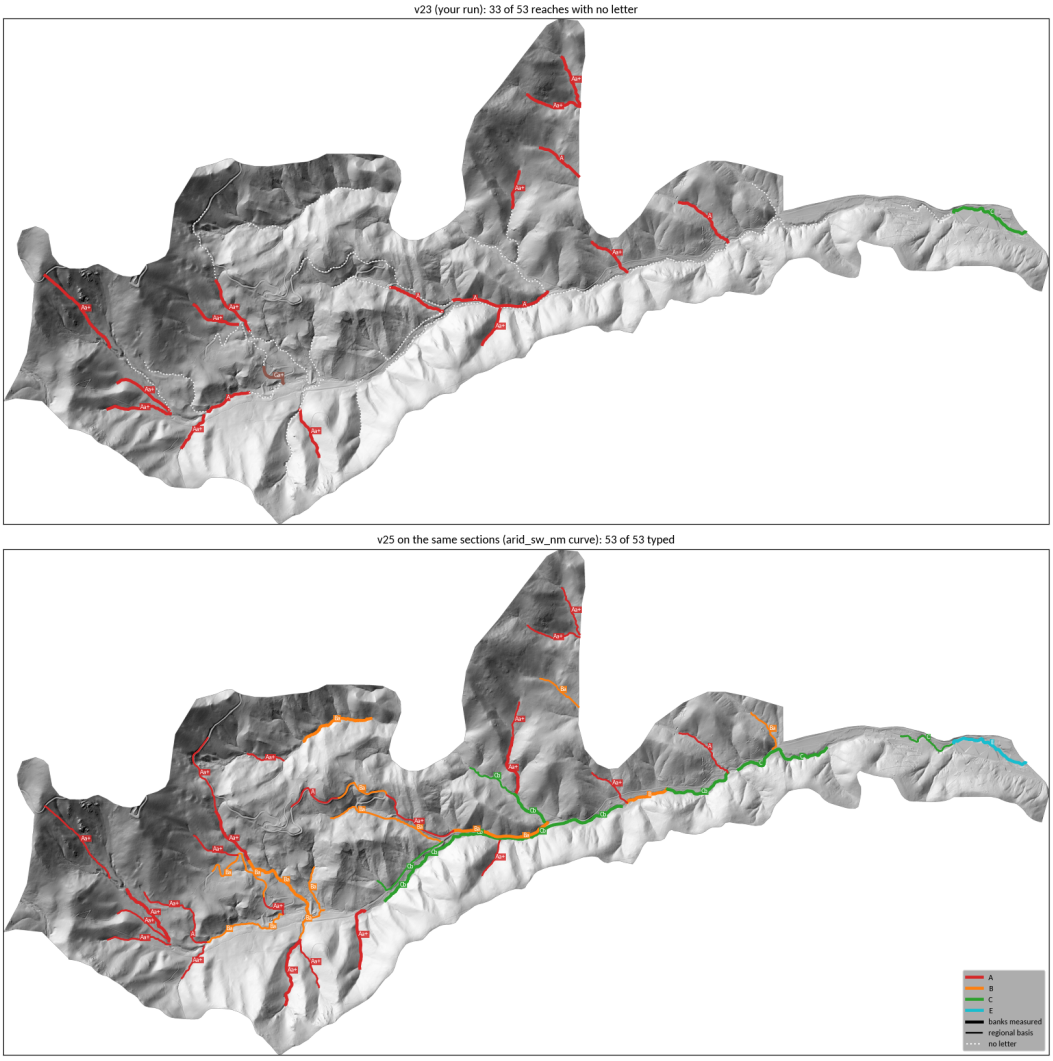


Figure 3. Stream types on Eagle Creek. Top: v23, 33 of 53 reaches without a letter (white dotted). Bottom: v0.25 on the same sections — all 53 typed: Aa+ 21, Ba 15, Cb 9, C 3, A 3, B 1, E 1. Thick lines rest on measured banks (20 reaches), thin on the regional channel (30) or one section (3). Steep headwaters are Aa+, the mid-slope tributaries Ba, and the main stem Cb/C/B.

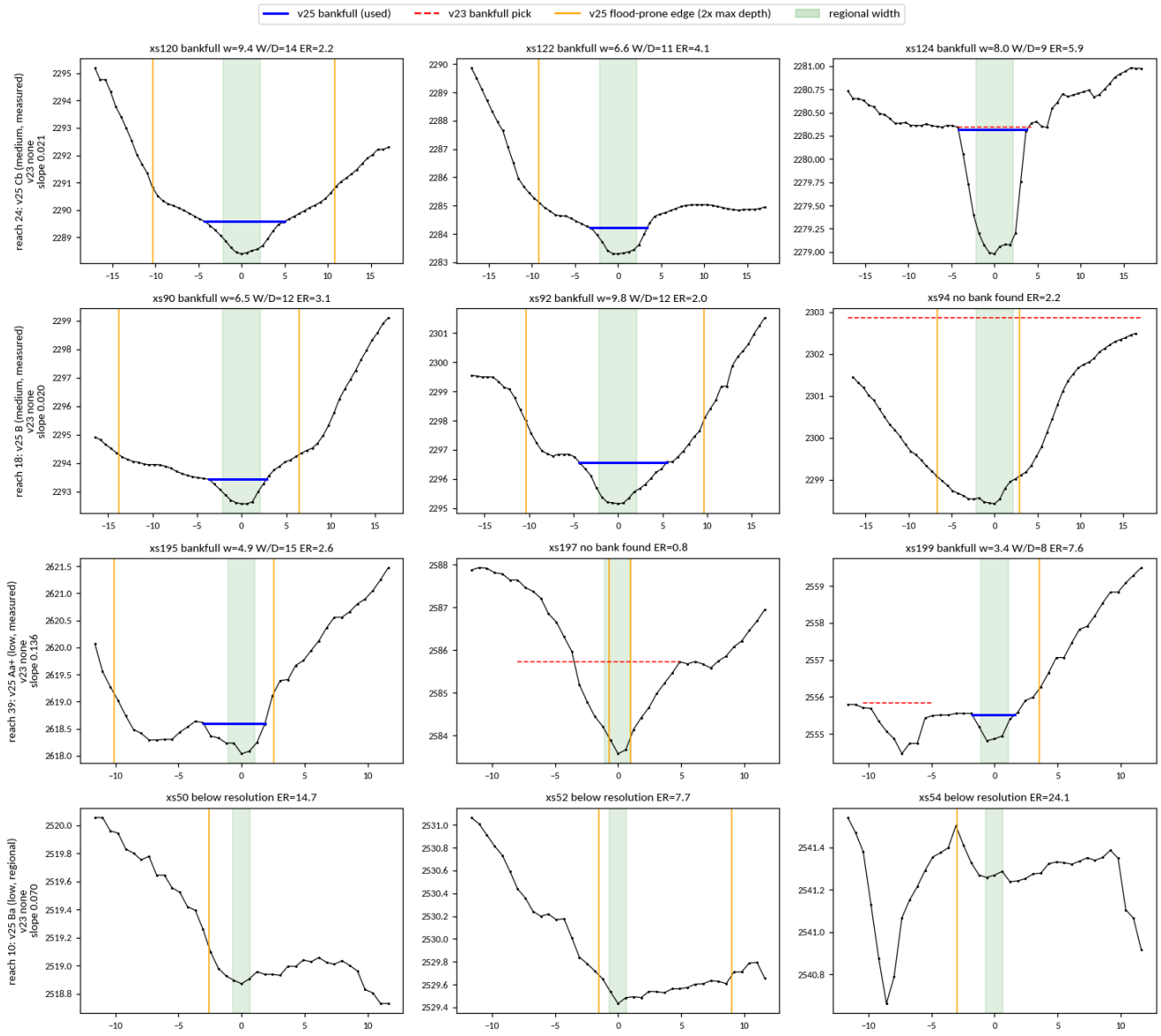


Figure 4. Cross-sections on four reaches (rows), three of the five on each. Blue: v0.25 bankfull where used; red dashed: the v23 pick; orange: v0.25 flood-prone edges; green band: the regional width. Rows 1–2 are the main stem, measured. Row 3 is a 13.6% tributary of 0.77 km^2, typed Aa+ with entrenchment 2.65; the middle section is a narrow V with no bank to find. Row 4 is a 7% tributary whose channel is under three samples wide; a flat bench beside it on every section — very likely a road — reads as floodplain.

No field data exist for this test area. Moody et al. (2003) list two field-typed reaches on a different Eagle Creek, near Alto, NM (below the South Fork: B4, entrenchment 1.9, width/depth 21.5, 2.2%; Tributary #1: A4a+, entrenchment 2.6, 12.7%); their published measurements key to their field types, which is a test of the key, not of geolens's measurements. The widths are the weaker part: main-stem picks here ran about twice the New Mexico curve's width, which led to the change in Section 5.2.

8 What to trust

Component	Status
Flow routing (pysheds)	Verified; WhiteboxTools not yet compared
Bank and floodplain delineation (reach scale)	Verified on known geometry
Bankfull (stream types)	Regional-stage method (v0.28) verified on known geometry; the old picker ran 2–2.6× Moody's field depths on Ruidoso. Neither compared with surveyed bankfull on these areas
Rosgen stream types	Key logic tested exhaustively (no label outside Rosgen's chart over 13,600 input combinations); reproduces Moody et al.'s two published field types (their Eagle Creek near Alto, NM). Screening only
Valley bottoms, confinement	Working, uncalibrated; the class breaks are defaults
Valley form and valley types	Measurements verified on valleys of known geometry; type thresholds encode Rosgen's descriptions and are not field-validated
Geology, NAIP	Both working on live services on Sam's machine
Gullies, headcuts, fans	Verified on fixtures
Units	The landscape pass, both type classifications, gullies and riparian compared on the same ground in feet and in metres; stream-type measurement also in feet with metre elevations. Headcuts, fans and the reach-scale path not yet. Fixing this test in v0.25 exposed a 3.28× error in valley width and confinement ratio on foot grids (v0.21–0.24), now corrected

Limitations to keep in mind

- **Bankfull leans on the regional curve.** Moody et al. (2003) note that Southwest bankfull is "rarely identified with the top of a bank", and the DEM picker found tops of banks; the regional stage fixes the depth instead, so a channel genuinely deeper or shallower than the curve (an incised arroyo, an E channel in a meadow) is measured at the wrong stage. Compare with surveyed sections where you have them. Lidar also returns the water surface, not the bed, on a perennial reach.
- **Roads.** A road bench beside a small channel looks like a floodplain: high entrenchment, a B or C where the channel may be an A. Check *stream_type_banks.gpkg* on a hillshade where a steep reach comes back B or C.
- **Small channels on coarse grids.** A channel under three samples wide is typed on the regional basis. On a 10 m DEM that is every channel; pass DEM_FINE.
- **The regional curve is regional.** Moody et al. recommend a local calibration curve for each watershed; a few surveyed sections would tighten every regional-basis call.
- **Geology cannot resolve narrow valleys** (source maps 1:50,000–1:1,000,000), and valley types that depend on it inherit that.
- **NDVI cannot tell alfalfa from cottonwood.** Fields touching the corridor merge into it; radar flood maps fail on dry sand beds.

9 Running it

Install once with *install_windows.bat* (or *bash install.sh*), which builds a permanent conda environment and checks itself on a synthetic valley; later versions update in place with *python update.py*. In the notebook, edit one cell:

```
DEM = r"C:\...\dem_3m.tif" # routing: 2-5 m is plenty
DEM_FINE = r"C:\...\DEM\Ruidoso_DEM_1m.tif" # channel sections, native lidar
OUT_DIR = r"C:\...\outputs\v25"
GEOLOGY = "auto"
REGIONAL_CURVE = "arid_sw_nm"
```

or call the API directly:

```
from geolens import api
api.plan_landscape_run(dem) # will it fit?
r = api.landscape_pass(dem, dem_fine=dem_1m, geology="auto",
                       regional_curve="arid_sw_nm")
r["stream_types"]["basis_counts"], r["stream_types"]["type_counts"]
```

The notebook checks the installed library's version and stops with the update command if it is older than the notebook. Read *RUNNING.md* for every parameter and *NEXT.md* for the current state of development.

References

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Data services: NAIP imagery from the USDA Farm Service Agency via Microsoft Planetary Computer; flood extents from the Copernicus Emergency Management Service Global Flood Monitoring product.